INCH-POUND

MIL-DTL-28840/28E
26 April 2007
SUPERSEDING
MIL-C-28840/28D(EC)
16 July 1986

DETAIL SPECIFICATION SHEET

# CONNECTORS, ELECTRICAL, CIRCULAR, SCREW THREADS, HIGH DENSITY, HIGH SHOCK, SHIPBOARD, CRIMP CONTACTS, PLUG ASSEMBLY, 90º, CLASSES DJ AND DJS

This specification is approved for use by all Departments and Agencies of the Department of Defense

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-28840.

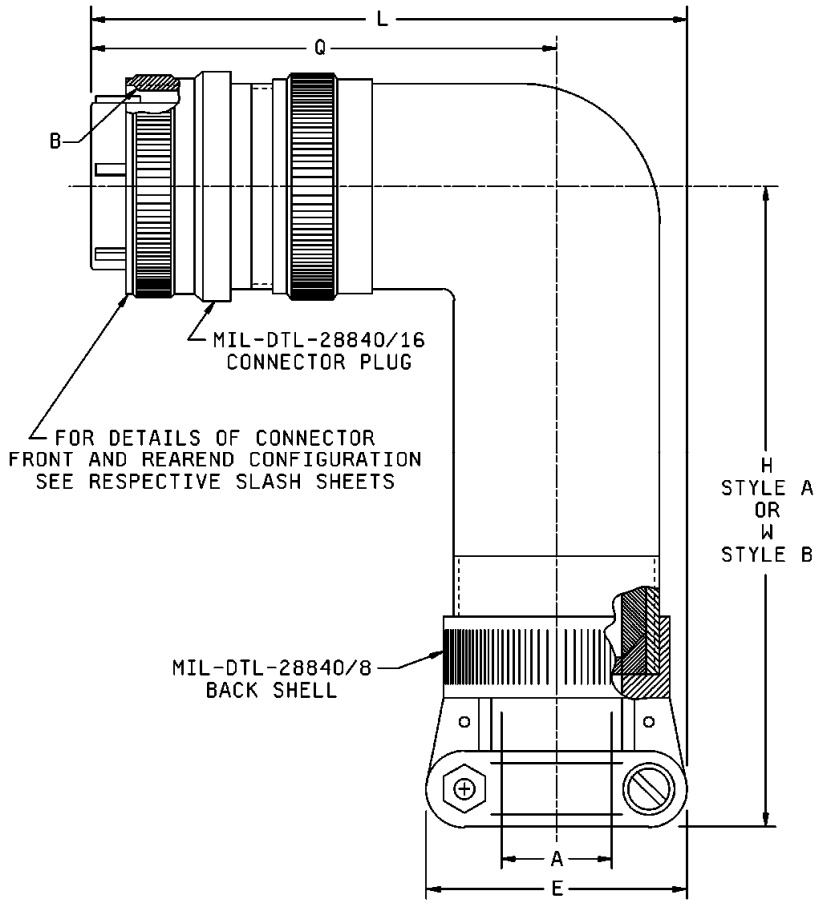



FIGURE 1. Dimensions and configuration.

AMSC N/A

FSC 5935

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Check the source to verify that this is the current version before use.

| Designator (See note 3) | Assembly number (See note 6) | Connector plug number (See note 6) | Backshell assembly number (See note 6) | A Cable entry | | B thread class 2B | E Ref |
|---|---|---|---|---|---|---|---|
| | | | | max | min | | |
| A(11) | M28840/28*A*AA** | M28840/16*A*** | M28840/801*A | .312 (7.92) | .188 (4.77) | .750-.1P -.2L-D.S. | .938 (23.82) |
| A(11) | M28840/28*A*AB** | M28840/16*A*** | M28840/801*B | | | | |
| A(11) | M28840/28*A*BA** | M28840/16*A*** | M28840/824*A | .438 (11.13) | .281 (7.14) | .750-.1P -.2L-D.S. | 1.125 (28.57) |
| A(11) | M28840/28*A*BB** | M28840/16*A*** | M28840/824*B | | | | |
| A(11) | M28840/28*A*CA** | M28840/16*A*** | M28840/825*A | .562 (14.27) | .344 (8.74) | .750-.1P -.2L-D.S. | 1.312 (33.32) |
| A(11) | M28840/28*A*CB** | M28840/16*A*** | M28840/825*B | | | | |
| B(13) | M28840/28*B*AA** | M28840/16*B*** | M28840/802*A | .312 (7.92) | .188 (4.77) | .875-.1P -.2L-D.S. | .938 (23.82) |
| B(13) | M28840/28*B*AB** | M28840/16*B*** | M28840/802*B | | | | |
| B(13) | M28840/28*B*BA** | M28840/16*B*** | M28840/803*A | .438 (11.13) | .281 (7.14) | .875-.1P -.2L-D.S. | 1.125 (28.57) |
| B(13) | M28840/28*B*BB** | M28840/16*B*** | M28840/803*B | | | | |
| B(13) | M28840/28*B*CA** | M28840/16*B*** | M28840/826*A | .562 (14.27) | .344 (8.74) | .875-.1P -.2L-D.S. | 1.312 (33.32) |
| B(13) | M28840/28*B*CB** | M28840/16*B*** | M28840/826*B | | | | |
| B(13) | M28840/28*B*DA** | M28840/16*B*** | M28840/827*A | .625 (15.87) | .375 (9.52) | .875-.1P -.2L-D.S. | 1.312 (33.32) |
| B(13) | M28840/28*B*DB** | M28840/16*B*** | M28840/827*B | | | | |
| C(15) | M28840/28*C*AA** | M28840/16*C*** | M28840/805*A | .438 (11.13) | .281 (7.14) | 1.062-.1P -.2L-D.S. | 1.125 (28.57) |
| C(15) | M28840/28*C*AB** | M28840/16*C*** | M28840/805*B | | | | |
| C(15) | M28840/28*C*BA** | M28840/16*C*** | M28840/806*A | .562 (14.27) | .344 (8.74) | 1.062-.1P -.2L-D.S. | 1.312 (33.32) |
| C(15) | M28840/28*C*BB** | M28840/16*C*** | M28840/806*B | | | | |
| C(15) | M28840/28*C*CA** | M28840/16*C*** | M28840/828*A | .625 (15.87) | .375 (9.52) | 1.062-.1P -.2L-D.S. | 1.312 (33.32) |
| C(15) | M28840/28*C*CB** | M28840/16*C*** | M28840/828*B | | | | |
| C(15) | M28840/28*C*DA** | M28840/16*C*** | M28840/807*A | .750 (19.05) | .438 (11.12) | 1.062-.1P -.2L-D.S. | 1.593 (40.46) |
| C(15) | M28840/28*C*DB** | M28840/16*C*** | M28840/807*B | | | | |
| C(15) | M28840/28*C*EA** | M28840/16*C*** | M28840/804*A | .312 (7.92) | .188 (4.77) | 1.062-.1P -.2L-D.S. | .938 (23.82) |
| C(15) | M28840/28*C*EB** | M28840/16*C*** | M28840/804*B | | | | |

FIGURE 1. Dimensions and configuration – Continued.



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Check the source to verify that this is the current version before use.

| Designator (See note 3) | Assembly number (See note 6) | Connector plug number (See note 6) | Backshell assembly number (See note 6) | H max Style A | N max Style B | L max | Q max |
|---|---|---|---|---|---|---|---|
| A(11) | M28840/28*A*AA** | M28840/16*A*** | M28840/801*A | 5.332 (135.43) | -- | 4.139 (105.13) | 3.306 (83.97) |
| A(11) | M28840/28*A*AB** | M28840/16*A*** | M28840/801*B | -- | 2.962 (75.23) | | |
| A(11) | M28840/28*A*BA** | M28840/16*A*** | M28840/824*A | 5.332 (135.43) | -- | | 3.306 (83.97) |
| A(11) | M28840/28*A*BB** | M28840/16*A*** | M28840/824*B | -- | 2.962 (75.23) | | |
| A(11) | M28840/28*A*CA** | M28840/16*A*** | M28840/825*A | 5.457 (138.61) | -- | 4.389 (111.48) | 3.431 (87.15) |
| A(11) | M28840/28*A*CB** | M28840/16*A*** | M28840/825*B | -- | 3.087 (78.41) | | |
| B(13) | M28840/28*B*AA** | M28840/16*B*** | M28840/802*A | 5.332 (135.43) | -- | 4.139 (105.13) | 3.306 (83.97) |
| B(13) | M28840/28*B*AB** | M28840/16*B*** | M28840/802*B | -- | 2.962 (75.23) | | |
| B(13) | M28840/28*B*BA** | M28840/16*B*** | M28840/803*A | 5.332 (135.43) | -- | 4.139 (105.13) | 3.306 (83.97) |
| B(13) | M28840/28*B*BB** | M28840/16*B*** | M28840/803*B | -- | 2.962 (75.23) | | |
| B(13) | M28840/28*B*CA** | M28840/16*B*** | M28840/826*A | 5.457 (138.61) | -- | 4.389 (111.48) | 3.431 (87.15) |
| B(13) | M28840/28*B*CB** | M28840/16*B*** | M28840/826*B | -- | 3.087 (78.41) | | |
| B(13) | M28840/28*B*DA** | M28840/16*B*** | M28840/827*A | 5.682 (144.32) | -- | 4.639 (117.83) | 3.556 (90.32) |
| B(13) | M28840/28*B*DB** | M28840/16*B*** | M28840/827*B | -- | 3.312 (84.12) | | |
| C(15) | M28840/28*C*AA** | M28840/16*C*** | M28840/805*A | 5.457 (138.61) | -- | 4.389 (111.48) | 3.431 (87.15) |
| C(15) | M28840/28*C*AB** | M28840/16*C*** | M28840/805*B | -- | 3.087 (78.41) | | |
| C(15) | M28840/28*C*BA** | M28840/16*C*** | M28840/806*A | 5.457 (138.61) | -- | | 3.431 (87.15) |
| C(15) | M28840/28*C*BB** | M28840/16*C*** | M28840/806*B | -- | 3.087 (78.41) | | |
| C(15) | M28840/28*C*CA** | M28840/16*C*** | M28840/828*A | 5.682 (144.32) | -- | 4.639 (117.83) | 3.556 (90.32) |
| C(15) | M28840/28*C*CB** | M28840/16*C*** | M28840/828*B | -- | 3.312 (84.12) | | |
| C(15) | M28840/28*C*DA** | M28840/16*C*** | M28840/807*A | 5.713 (145.11) | -- | | 3.556 (90.32) |
| C(15) | M28840/28*C*DB** | M28840/16*C*** | M28840/807*B | -- | 3.343 (84.91) | | |
| C(15) | M28840/28*C*EA** | M28840/16*C*** | M28840/804*A | 5.457 (138.61) | -- | 4.389 (111.48) | 3.431 (87.15) |
| C(15) | M28840/28*C*EB** | M28840/16*C*** | M28840/804*B | -- | 3.087 (78.41) | | |

FIGURE 1. Dimensions and configuration – Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:45Z
Check the source to verify that this is the current version before use.

| Designator (See note 3) | Assembly number (See note 6) | Connector plug number (See note 6) | Backshell assembly number (See note 6) | A Cable entry max | A Cable entry min | B thread class 2B | E Ref |
|---|---|---|---|---|---|---|---|
| D(17) | M28840/28*D*AA** | M28840/16*D*** | M28840/808*A | .625 (15.87) | .375 (9.52) | 1.125-.1P -.2L-D.S. | 1.312 (33.32) |
| D(17) | M28840/28*D*AB** | M28840/16*D*** | M28840/808*B | | | | |
| D(17) | M28840/28*D*BA** | M28840/16*D*** | M28840/809*A | .750 (19.05) | .438 (11.12) | 1.125-.1P -.2L-D.S. | 1.593 (40.46) |
| D(17) | M28840/28*D*BB** | M28840/16*D*** | M28840/809*B | | | | |
| D(17) | M28840/28*D*CA** | M28840/16*D*** | M28840/810*A | .938 (23.82) | .625 (15.87) | 1.125-.1P -.2L-D.S. | 1.750 (44.45) |
| D(17) | M28840/28*D*CB** | M28840/16*D*** | M28840/810*B | | | | |
| E(19) | M28840/28*E*AA** | M28840/16*E*** | M28840/811*A | .625 (15.87) | .375 (9.52) | 1.312-.1P -.2L-D.S. | 1.312 (33.32) |
| E(19) | M28840/28*E*AB** | M28840/16*E*** | M28840/811*B | | | | |
| E(19) | M28840/28*E*BA** | M28840/16*E*** | M28840/812*A | .938 (23.82) | .625 (15.87) | 1.312-.1P -.2L-D.S. | 1.750 (44.45) |
| E(19) | M28840/28*E*BB** | M28840/16*E*** | M28840/812*B | | | | |
| F(23) | M28840/28*F*AA** | M28840/16*F*** | M28840/813*A | .750 (19.05) | .438 (11.12) | 1.500-.1P -.2L-D.S. | 1.593 (40.46) |
| F(23) | M28840/28*F*AB** | M28840/16*F*** | M28840/813*B | | | | |
| F(23) | M28840/28*F*BA** | M28840/16*F*** | M28840/814*A | .938 (23.82) | .625 (15.87) | 1.500-.1P -.2L-D.S. | 1.750 (44.45) |
| F(23) | M28840/28*F*BB** | M28840/16*F*** | M28840/814*B | | | | |
| F(25) | M28840/28*F*CA** | M28840/16*F*** | M28840/815*A | 1.188 (30.18) | .812 (20.62) | 1.500-.1P -.2L-D.S. | 2.093 (53.16) |
| F(25) | M28840/28*F*CB** | M28840/16*F*** | M28840/815*B | | | | |

FIGURE 1. Dimensions and configuration – Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:45Z
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| Designator (See note 3) | Assembly number (See note 6) | Connector plug number (See note 6) | Backshell assembly number (See note 6) | H max Style A | N max Style B | L max | Q max |
|---|---|---|---|---|---|---|---|
| D(17) | M28840/28*D*AA** | M28840/16*D*** | M28840/808*A | 5.682 (144.32) | -- | 4.639 (117.83) | 3.556 (90.32) |
| D(17) | M28840/28*D*AB** | M28840/16*D*** | M28840/808*B | -- | 3.312 (84.12) | | |
| D(17) | M28840/28*D*BA** | M28840/16*D*** | M28840/809*A | 5.713 (145.11) | -- | | 3.556 (90.32) |
| D(17) | M28840/28*D*BB** | M28840/16*D*** | M28840/809*B | -- | 3.343 (84.91) | | |
| D(17) | M28840/28*D*CA** | M28840/16*D*** | M28840/810*A | 5.807 (147.50) | -- | | 3.556 (90.32) |
| D(17) | M28840/28*D*CB** | M28840/16*D*** | M28840/810*B | -- | 3.437 (87.30) | | |
| E(19) | M28840/28*E*AA** | M28840/16*E*** | M28840/811*A | 5.682 (144.32) | -- | | 3.556 (90.32) |
| E(19) | M28840/28*E*AB** | M28840/16*E*** | M28840/811*B | -- | 3.312 (84.12) | | |
| E(19) | M28840/28*E*BA** | M28840/16*E*** | M28840/812*A | 5.832 (148.13) | -- | 4.889 (124.18) | 3.681 (93.50) |
| E(19) | M28840/28*E*BB** | M28840/16*E*** | M28840/812*B | -- | 3.462 (87.93) | | |
| F(23) | M28840/28*F*AA** | M28840/16*F*** | M28840/813*A | 5.738 (145.75) | -- | | 3.681 (93.50) |
| F(23) | M28840/28*F*AB** | M28840/16*F*** | M28840/813*B | -- | 3.368 (85.55) | | |
| F(23) | M28840/28*F*BA** | M28840/16*F*** | M28840/814*A | 5.832 (148.13) | -- | | 3.681 (93.50) |
| F(23) | M28840/28*F*BB** | M28840/16*F*** | M28840/814*B | -- | 3.462 (87.93) | | |
| F(25) | M28840/28*F*CA** | M28840/16*F*** | M28840/815*A | 6.019 (152.88) | -- | | 3.681 (93.50) |
| F(25) | M28840/28*F*CB** | M28840/16*F*** | M28840/815*B | -- | 3.649 (92.68) | | |

FIGURE 1. Dimensions and configuration – Continued.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T19:45Z
Check the source to verify that this is the current version before use.

| Designator (See note 3) | Assembly number (See note 6) | Connector plug number (See note 6) | Backshell assembly number (See note 6) | A Cable entry | | B thread class 2B | E Ref |
|---|---|---|---|---|---|---|---|
| | | | | max | min | | |
| G(25) | M28840/28*G*AA** | M28840/16*G*** | M28840/816*A | .938 (23.82) | .625 (15.87) | 1.625-.1P -.2L-D.S. | 1.750 (44.45) |
| G(25) | M28840/28*G*AB** | M28840/16*G*** | M28840/816*B | | | | |
| G(25) | M28840/28*G*BA** | M28840/16*G*** | M28840/817*A | 1.188 (30.18) | .812 (20.62) | 1.625-.1P -.2L-D.S. | 2.093 (53.16) |
| G(25) | M28840/28*G*BB** | M28840/16*G*** | M28840/817*B | | | | |
| G(25) | M28840/28*G*CA** | M28840/16*G*** | M28840/818*A | 1.250 (31.75) | .875 (22.22) | 1.625-.1P -.2L-D.S. | 2.093 (53.16) |
| G(25) | M28840/28*G*CB** | M28840/16*G*** | M28840/818*B | | | | |
| G(25) | M28840/28*G*DA** | M28840/16*G*** | M28840/819*A | 1.380 (35.05) | 1.000 (25.40) | 1.625-.1P -.2L-D.S. | 2.343 (59.51) |
| G(25) | M28840/28*G*DB** | M28840/16*G*** | M28840/819*B | | | | |
| H(29) | M28840/28*H*AA** | M28840/16*H*** | M28840/820*A | 1.188 (30.18) | .812 (20.62) | 1.812-.1P -.2L-D.S. | 2.093 (53.16) |
| H(29) | M28840/28*H*AB** | M28840/16*H*** | M28840/820*B | | | | |
| H(29) | M28840/28*H*BA** | M28840/16*H*** | M28840/821*A | 1.380 (35.05) | 1.000 (25.40) | 1.812-.1P -.2L-D.S. | 2.343 (59.51) |
| H(29) | M28840/28*H*BB** | M28840/16*H*** | M28840/821*B | | | | |
| J(33) | M28840/28*J*AA** | M28840/16*J*** | M28840/822*A | 1.380 (35.05) | 1.000 (25.40) | 2.000-.1P -.2L-D.S. | 2.343 (59.51) |
| J(33) | M28840/28*J*AB** | M28840/16*J*** | M28840/822*B | | | | |
| J(33) | M28840/28*J*BA** | M28840/16*J*** | M28840/823*A | 1.625 (41.27) | 1.250 (31.75) | 2.000-.1P -.2L-D.S. | 2.750 (69.85) |
| J(33) | M28840/28*J*BB** | M28840/16*J*** | M28840/823*B | | | | |

FIGURE 1. Dimensions and configuration – Continued.



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| Designator (See note 3) | Assembly number (See note 6) | Connector plug number (See note 6) | Backshell assembly number (See note 6) | H max Style A | N max Style B | L max | Q max |
|---|---|---|---|---|---|---|---|
| G(25) | M28840/28*G*AA** | M28840/16*G*** | M28840/816*A | 5.832 (148.13) | -- | 4.889 (124.18) | 3.681 (93.50) |
| G(25) | M28840/28*G*AB** | M28840/16*G*** | M28840/816*B | -- | 3.462 (87.93) | | |
| G(25) | M28840/28*G*BA** | M28840/16*G*** | M28840/817*A | 6.019 (152.88) | -- | | 3.681 (93.50) |
| G(25) | M28840/28*G*BB** | M28840/16*G*** | M28840/817*B | -- | 3.649 (92.68) | | |
| G(25) | M28840/28*G*CA** | M28840/16*G*** | M28840/818*A | 6.144 (156.06) | -- | 5.139 (130.53) | 3.806 (96.67) |
| G(25) | M28840/28*G*CB** | M28840/16*G*** | M28840/818*B | -- | 3.774 (95.86) | | |
| G(25) | M28840/28*G*DA** | M28840/16*G*** | M28840/819*A | 6.176 (156.87) | -- | | 3.806 (96.67) |
| G(25) | M28840/28*G*DB** | M28840/16*G*** | M28840/819*B | -- | 3.806 (96.67) | | |
| H(29) | M28840/28*H*AA** | M28840/16*H*** | M28840/820*A | 6.144 (156.06) | -- | | 3.806 (96.67) |
| H(29) | M28840/28*H*AB** | M28840/16*H*** | M28840/820*B | -- | 3.774 (95.86) | | |
| H(29) | M28840/28*H*BA** | M28840/16*H*** | M28840/821*A | 6.176 (156.87) | -- | | 3.806 (96.67) |
| H(29) | M28840/28*H*BB** | M28840/16*H*** | M28840/821*B | -- | 3.806 (96.67) | | |
| J(33) | M28840/28*J*AA** | M28840/16*J*** | M28840/822*A | 6.426 (163.22) | -- | 5.639 (143.23) | 4.056 (103.02) |
| J(33) | M28840/28*J*AB** | M28840/16*J*** | M28840/822*B | -- | 4.056 (103.02) | | |
| J(33) | M28840/28*J*BA** | M28840/16*J*** | M28840/823*A | 6.701 (170.21) | -- | | 4.056 (103.02) |
| J(33) | M28840/28*J*BB** | M28840/16*J*** | M28840/823*B | -- | 4.331 (110.01) | | |

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Shell sizes are provided within parentheses for information and are not a part of the designator.
4. Dimensions apply after plating.
5. Normal mating key position shown. Alternate key positions in accordance with MIL-DTL-28840.
6. Assembly shall be made of a connector and backshell of the same material.

FIGURE 1. Dimensions and configuration – Continued.

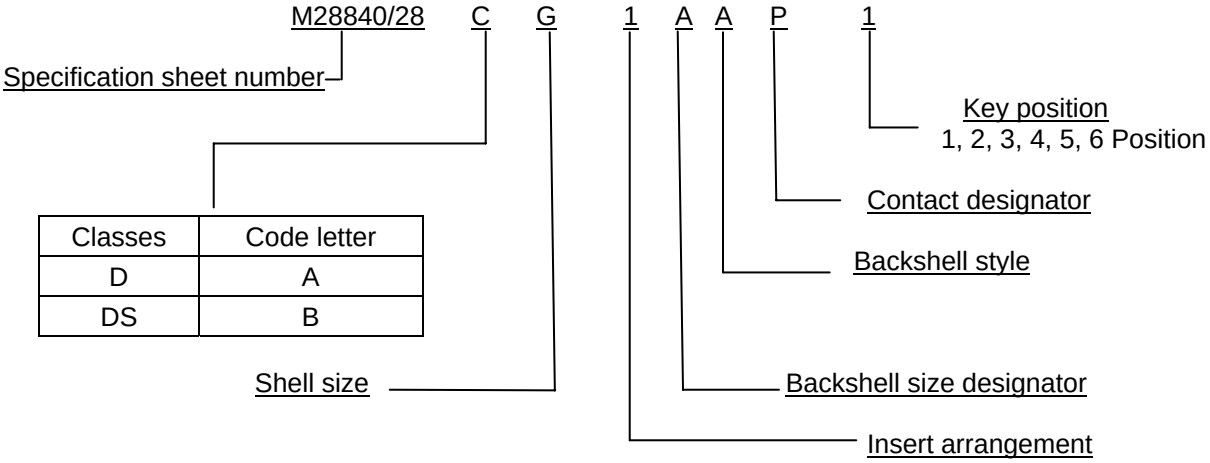


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REQUIREMENTS:

Dimensions and configuration: See figure 1.
Classes: DJ and DJS.
Insert Arrangements: See the section of MIL-STD-1698 corresponding to desired shell size.
Contacts: Style P or S crimp type.
Shell polarization: 1 thru 6, key position.
Marking: Part or Identifying Number (PIN) on shell.
Mating counterpart: Mating connectors in accordance with MIL-DTL-28840/10, 11, 12, 14, 20 and 21.
Installation and removal tool: In accordance MIL-I-81969/33 and /34.
PIN's shown indicate aluminum alloy hardware. If corrosion resistant steel parts are required, the connector and accessory part numbers must be changed to designate corrosion resistant hardware.

PIN example:




| Classes | Code letter |
|---|---|
| D | A |
| DS | B |

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Referenced documents. In addition to MIL-DTL-28840, this document references the following:

MIL-DTL-28840/8
MIL-DTL-28840/10
MIL-DTL-28840/11
MIL-DTL-28840/12
MIL-DTL-28840/14
MIL-DTL-28840/16
MIL-DTL-28840/20
MIL-DTL-28840/21
MIL-I-81969/33
MIL-I-81969/34
MIL-STD-1698



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CONCLUDING MATERIAL

Custodians:
Navy - EC
DLA - CC

Preparing activity:
DLA - CC

(Project 5935–2005-043)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.



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